Software Engineering and Data Science
SEIS 766: Vision AI, Assignment #7

Due Date: March 26th

This assignment uses three (3) datasets: "**CellDNA_data1**.csv", "**CellDNA_data2**.csv", and "**CellDNA_targets**.csv". "**CellDNA_data1**.csv" and "**CellDNA_data2**.csv" contains 6 and 7 predictors, respectively. "**CellDNA_targets**.csv" contains the target Ys for all the 1217 records . Write a program using either Python+Keras or MatLab to perform a **two-class** classification on ****ALL**** the records in this dataset.

1. Read in data ***X*** from "**CellDNA_data1**.csv" and "**CellDNA_data2**.csv". Remember performing standardization separately on each dataset your program reads in from each file.
2. Convert the target dependent variable ***Ys*** in "**CellDNA_targets**.csv" to **binary values** of either 0s or 1s for your **two-class** classification.
3. Build your regular neural network that takes in inputs from two data sources of "**CellDNA_data1**.csv" and "**CellDNA_data2**.csv".
4. After taking in data from each source, perform few dense (fully-connect) layers transformation similar to the NN architecture shown in "VAI_AE_2_VAE_FAPI_TrainLoop_**25F**.pdf".
5. Your NN then merge / concatenate two ***Z***s from each stream you built in the previous step.
6. Finally, your NN performs few more dense (fully-connect) layers transformation for final classification. **Display** your network architecture.
7. Train your NN using "**fit**" or "**trainNetwork**" and show the confusion matrix generated from this training.
8. Train your NN AGAIN using **your own** "**training loop**" and **your own** "**binary cross entropy**" loss function. Show the confusion matrix from this training. In this step, you ar ****NOT****allowed to train your NN using the "**fit**" or "**trainNetwork**" function.

Submission Guideline:

1. Please include the answers to the above questions in a WORD document.
2. Please print your program (matlab or python) as **PDF / html** and include the **PDF / html** in your submission.
3. Please include your program "a6.m/.mlx/.py/.inpyb" in your submission.
4. Prepare EVERYTHING mentioned in the guideline and submit them on **Canvas** no later than the due date. Please do **NOT** zip your files.
5. Please carefully follow the submission guideline. Otherwise, the instructor may not be able to grade your assignment.